

Day 43



Static Method in JavaScript:

What is a static method?

- A static method is a function defined on the class itself, not on the objects created from the class.
- You call it using the class name, not the object.

Key points:

- You define them using the keyword static.
- You call them using the class name, not the object.
- They are useful for utility functions that don't depend on instance data.

When to use static methods?

Use a static method when:

- The method does not depend on instance-specific data
- It's a utility/helper function
- You need a way to create or manage instances

Examples:

- Math.random()
- Date.now()

Example: a very basic example. You created a class, you put a static method in it, and then you called it.

```
JS main.js > ...
1 class Hello {
2   static sayHi() {
3     console.log("Hi!");
4   }
5 }
6
7 Hello.sayHi(); // prints "Hi!"
8
```

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```
PS E:\JavaScript\Day41-50\Day43> node main.js
Hi!
```

Important Note:

1. You must call a static method using the class name, like this:

```
Hello.sayHi();
```

2. You cannot call it from an object like this:

```
let h = new Hello();
```

```
h.sayHi(); // ❌ This will NOT work
```

Static means: belongs to the class, not to the object.

Example: static method with parameters

```
8
9  class MathHelper {
10    static add(a, b) {
11      return a + b;
12    }
13  }
14  console.log(MathHelper.add(5, 3)); // 8
```

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```
● PS E:\JavaScript\Day41-50\Day43> node main.js
8
```

- static add(a, b) is a static method.
- It takes two numbers and returns their sum.
- You call it using the class name, not an object.

Example: compare Static vs Non-Static

```
16  class Person {
17    constructor(name) {
18      this.name = name;
19    }
20    // Instance method (belongs to the object)
21    sayHello() {
22      console.log("Hello, I'm " + this.name);
23    }
24    // Static method (belongs to the class)
25    static sayWelcome() {
26      console.log("Welcome to the class!");
27    }
28  }
29  Person.sayWelcome(); // static method → works
30  let p = new Person("Alice");
31  p.sayHello();        // instance method → works
32  // p.sayWelcome();    // ❌ ERROR: object can't use static method
```

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```
PS E:\JavaScript\Day41-50\Day43> node main.js
Welcome to the class!
Hello, I'm Alice
```

--The End--